OOP Project Proposal

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App Name

- Auto Drive Dash

App Description

- ADD (Auto Drive Dash) is an endless side-scrolling game that focuses on players controlling a vehicle of their choosing through dynamically generated highways filled with obstacles, coins, and power-ups. The goal is to survive as long as possible while collecting rewards and avoiding collisions. The game is designed for casual players who enjoy fast-paced reflex-based games, providing an engaging experience that combines speed, precision, and strategy.

Major Features

Feature	**Description**
User Authentication	Players can register, log in, and save their progress and high scores securely.
Player Control	Players can move around and accelerate to avoid obstacles and collect powerups using keyboard controls which can be changed in settings.
Score & Progress Tracking	Keeps track of the player’s score, coins collected, and high scores during gameplay.
Dynamic Game System	Generates random enemy vehicles, power-ups, and music with more challenging gameplay as players reach higher scores.
Automatic Data Saving	Automatically stores player progress and high scores in a database after each session, allowing players to continue from where they left off without manual saving.

Technologies Used

Category	**Tools/Frameworks**
Programming Language	Java
Frameworks/Libraries	JavaFX FXGL MySQL Connector/J
Architecture	MVC (Model-View-Controller) ECS (Entity-Component-System)
Tools	Visual Studio Code, GitHub, SceneBuilder, Piskel, Canva, MySQL, Lucidchart

Reflection Report

Developing the Auto Drive Dash (ADD) project has allowed me to deeply reflect on how the four main principles of Object-Oriented Programming, encapsulation, abstraction, inheritance, and polymorphism, naturally shape and strengthen the structure of a real-world application. At first, the idea of building an endless side-scrolling driving game seemed focused mainly on visuals, controls, and fun gameplay mechanics. However, as I analyzed the proposal and began breaking it into components such as user authentication, player control, score tracking, dynamic obstacles, and data saving, I realized that OOP is not just a programming style but a way of thinking that organizes complexity into manageable, logical, and reusable parts. Each feature of the game is an opportunity to apply the four OOP principles in a way that improves code clarity, maintainability, and scalability.

Overall, the Auto Drive Dash project has strengthened my understanding of how the four principles of Object-Oriented Programming go beyond theory and become practical tools in system design. Encapsulation taught me the importance of data protection and organization. Abstraction helped me focus on functionality rather than complexity. Inheritance showed me how to reuse and structure code efficiently. Polymorphism revealed how flexibility and dynamic behavior can exist within a unified system. By consciously applying these principles, the proposed game becomes not only more functional and engaging for players, but also more professional, maintainable, and expandable as a software product. This reflection made me realize that mastering OOP is essential not only for creating better programs, but also for thinking more logically and systematically as a future software developer.